ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

TASK – 02

House Price Prediction using Machine Learning

1. **Introduction**

House price prediction is an important real-world problem in the real estate domain. Accurate prediction helps buyers, sellers, and investors make informed decisions. In this project, we built a machine learning model to predict house prices using various features from the California Housing dataset.

The main objective is to apply feature engineering, train multiple machine learning models, evaluate their performance, and select the best model.

2. **Dataset Description**

The dataset used is the California Housing Dataset, available in the Scikit-learn library.

Key Features:

- Median Income
- House Age
- Average Rooms
- Average Bedrooms
- Population
- Latitude & Longitude

Target Variable:

- **House Price (Median Value of Houses)**

The dataset is clean and does not contain missing values, making it suitable for model training.

3. **Data Preprocessing**

Data preprocessing is a crucial step in machine learning.

Steps performed:

- Checked dataset structure and data types
- Verified missing values
- Separated features (X) and target (y)

Feature Scaling:

Feature scaling was applied using StandardScaler to normalize the data. This ensures that all features contribute equally to model training and improves model performance.

4. **Model Building**

Three different regression models were used:

1. **Linear Regression**
   - Simple and interpretable model
   - Assumes linear relationship between variables
2. **Ridge Regression**
   - Improved version of Linear Regression
   - Helps reduce overfitting using regularization
3. **Decision Tree Regressor**
   - Captures non-linear relationships
   - Splits data into decision-based rules

5. **Model Evaluation**

Models were evaluated using the following metrics:

RMSE (Root Mean Squared Error)

- Measures prediction error
- Lower value indicates better performance

R^2 Score (Coefficient of Determination)

- Measures how well the model explains variance
- Higher value indicates better performance

Each model was trained and tested, and predictions were compared.

6. **Results & Comparison**

A comparison table was created to analyze model performance based on RMSE and R^2 score.

Observations:

- Linear Regression provides baseline performance
- Ridge Regression reduces overfitting slightly

- Decision Tree captures complex patterns better
- The best model was selected based on:
  ✓ Lowest RMSE
  ✓ Highest R^2 Score

7. Visualization

A scatter plot of Actual vs Predicted values was created to evaluate model performance visually.

- Points close to the diagonal line indicate accurate predictions
- More deviation indicates higher error

This helps in understanding how well the model generalizes.

8. Conclusion

This project successfully demonstrated an end-to-end machine learning pipeline:

- Data preprocessing and scaling
- Training multiple models
- Evaluating performance
- Selecting the best model

The final model provides a reliable way to predict house prices and can be further improved with advanced techniques like hyperparameter tuning and ensemble methods.

9. Key Learnings

- Importance of feature scaling
- Comparing multiple models is essential
- Understanding evaluation metrics like RMSE and R^2
- Practical implementation of machine learning workflow